AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2011

Physics Paper I

Time allowed: 35 minutes Marks 25

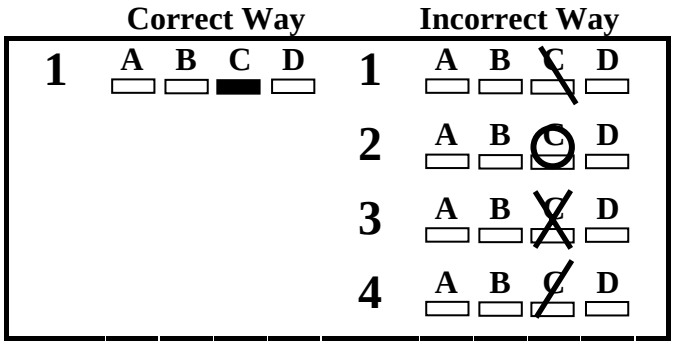
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.

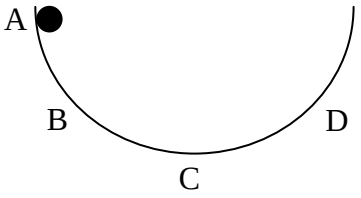


5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. In the cross section of a bowl, a metallic ball executed simple harmonic motion. If the ball starts its motion from point A, then at which point will it have the maximum kinetic energy?



2. Which of the following produces longitudinal waves in the particles of a medium?

   A. Vibration in a rope
   B. To and fro motion in a spring
   C. Vibration on the surface of water
   D. To and fro motion of a bob in a simple pendulum

3. Loudness of sound does NOT depend upon

   A. area of vibrating body.
   B. amplitude of vibrating body.
   C. distance from vibrating body.
   D. time period of vibrating body.

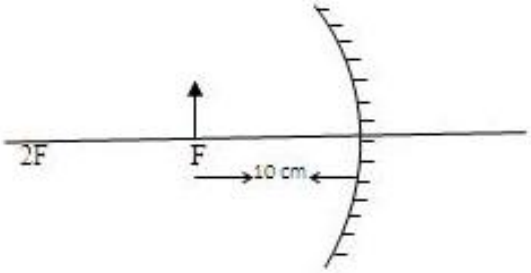
4. Ultrasound is used for different purposes. Which of the following is NOT a current use of ultrasound?

   A. Detection of fault in engine
   B. Measure the depth of an ocean
   C. Diagnosis of different diseases
   D. Ranging and detection of aeroplanes

5. Where should an object be placed to get an erect and virtual image by a concave mirror?

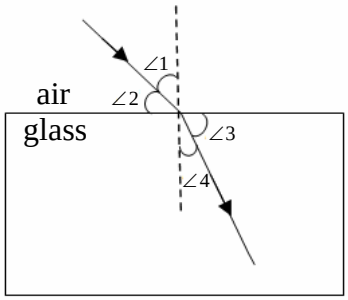
   A. At *F*
   B. Beyond *C*
   C. Between *F* and *P*
   D. Between *C* and *F*

6. The focal length of a concave mirror is 10 cm. An object of height 5 cm is lying at its focus. What is the image distance of the object?



   A. 0
   B. 5 cm
   C. 10 cm
   D. Can not be defined

7. In the given ray diagram, which pair of angles states angle of incidence and angle of refraction respectively?



A. $\angle 1$ and $\angle 3$
B. $\angle 1$ and $\angle 4$
C. $\angle 2$ and $\angle 3$
D. $\angle 2$ and $\angle 4$

8. The phenomenon due to which a pool of water appears less deep than actual is called

A. reflection.
B. refraction.
C. diffraction.
D. interference.

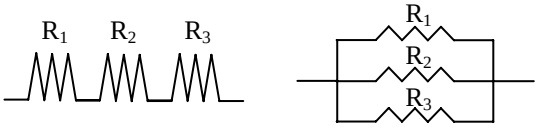
9. A hair comb becomes positively charged because

A. hair has extra positive charge.
B. neutrons were created in the comb.
C. the comb loses electrons to the hair.
D. hair acquires extra protons from the comb.

10. Which of the following properties of electric field lines is FALSE?

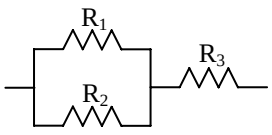
A. They are vector quantities.
B. They never intersect each other.
C. They try to follow the longest path.
D. They start from positive and terminate at negative charge.

11. Which of the following combinations of resistors is equivalent to the highest resistance produced in a circuit i.e. $(R_1 = R_2 = R_3 = 2\Omega)$?

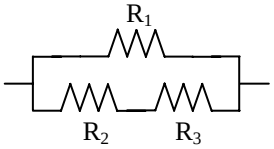


(A)

(B)



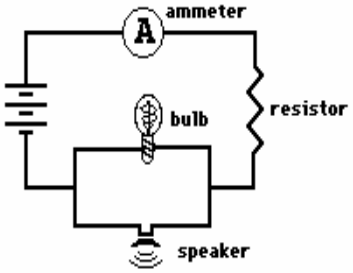
(C)



(D)

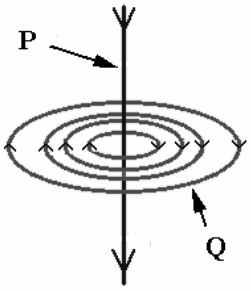
12. An electric circuit is shown below. Which of the two electrical devices is connected in series and in parallel?

	Series	Parallel
A	Speaker and bulb	Ammeter and resistor
B	Bulb and resistor	Ammeter and speaker
C	Ammeter and bulb	Speaker and resistor
D	Ammeter and resistor	Speaker and bulb

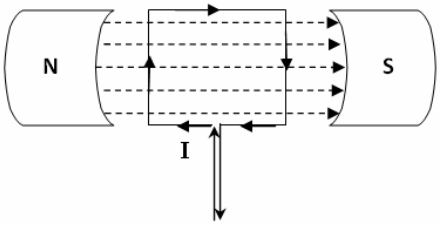


13. In the given figure, points 'P' and 'Q' represent

	P	Q
A	Current	Magnetic field
B	Current	Electric field
C	Magnetic field	Electric field
D	Electric field	Magnetic field



14. In the given diagram, a rectangular current carrying coil is lying between the poles of a magnet. The direction of motion in the coil will be



A. vertically upward.
B. clockwise rotation.
C. vertically downward.
D. anticlockwise rotation.

15. A DC motor works on the principle of



A. heating effect.
B. magnetic effect.
C. Compton's effect.
D. photoelectric effect.

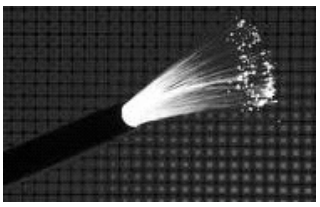
16. Which of the following devices is an example of electromagnetic induction?

A. DC motor
B. Electric bell
C. Galvanometer
D. AC generator

17. CRO has many uses in the medical field. Which of the following is NOT a correct use of it?

A. Use as a voltmeter
B. Use in studying the information of waves.
C. Use in studying the effect of electric field on moving charges.
D. Use in studying the effect of background radiation on living things.

18. Which of the following is NOT used to send information in the form of digital signals in optical fibres?

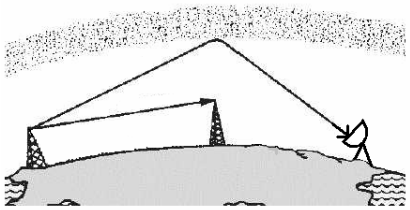


A. Photon
B. Neutron
C. Pulse of light
D. Quanta of light

19. On which of the following principles of light does an optical fibre work?

A. Reflection
B. Refraction
C. Diffraction
D. Total internal reflection

20. When a radio wave is intercepted by an antenna, the antenna converts the wave into



A. a digital signal.
B. an analog signal.
C. an electrical signal.
D. an electromagnetic signal.

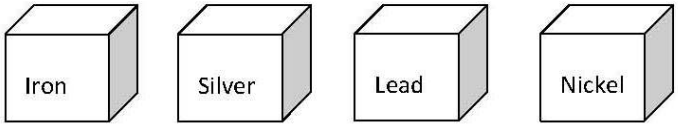
21. The nucleus of an element consists of 90 protons and 144 neutrons. After emitting two beta particles followed by an alpha particle, the nucleus will have

A. 86 protons and 140 neutrons.
B. 88 protons and 142 neutrons.
C. 90 protons and 140 neutrons.
D. 90 protons and 142 neutrons.

22. Atomic number of an atom is 56 and its mass number is 138. Which of the following is the correct number of electrons, protons and neutrons of this atom?

	Electrons	**Protons**	**Neutrons**
A	56	56	82
B	56	56	138
C	82	56	56
D	138	82	56

23. Which of the following boxes is used to keep radioactive material?

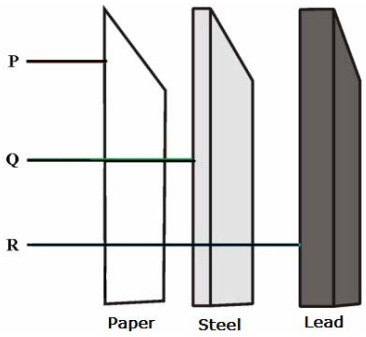


(A) (B) (C) (D)

24. Which of the following radiations bends in a magnetic field with largest radius?

A. Alpha particles
B. Beta particles
C. Gamma rays
D. X-rays

25. In the figure different sheets are placed in front of nuclear radiations. Which of the ray/s can be identified as gamma radiation?



A. P
B. Q
C. R
D. P and Q

END OF PAPER

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